

● EASY

● GEOMETRY AND TRIGONOMETRY

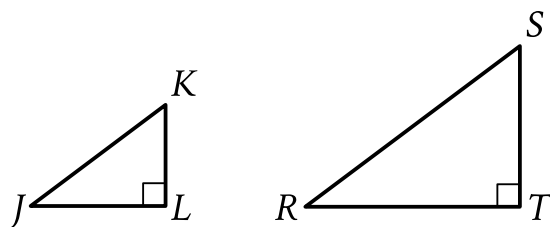
Geometry and Trigonometry

01

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY

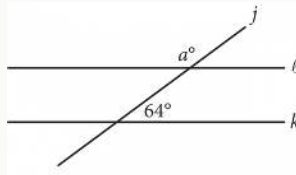


Note: Figure not drawn to scale.

- Right triangle upper J upper K upper L is labeled as follows:
 - Angle upper L is a right angle.
- Right triangle upper R upper S upper T is labeled as follows:
 - Angle upper T is a right angle.
- A note indicates the figures are not drawn to scale.

In the figure shown, triangle JKL is similar to triangle RST , where J corresponds to R and K corresponds to S . The length of $\bar{JK}$ is 15, and the perimeter of triangle JKL is 36. The length of $\bar{RS}$ is 135. What is the perimeter of triangle RST ?

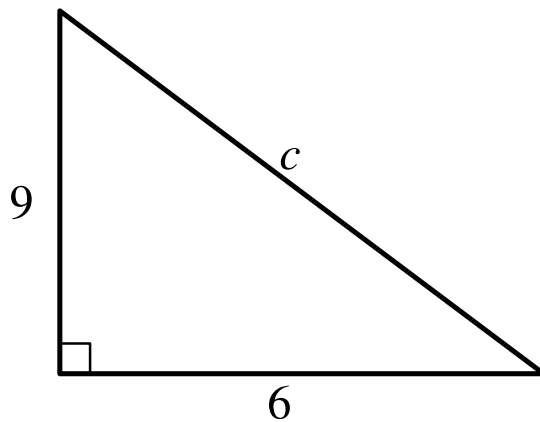
- A. 324
- B. 540
- C. 2,916
- D. 8,100



Note: Figure not drawn to scale.

In the figure above, lines ℓ and k are parallel. What is the value of a ?

- A. 26
- B. 64
- C. 116
- D. 154

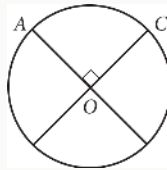


Note: Figure not drawn to scale.

- One angle is a right angle.
- The length of the side opposite the right angle is c .
- The lengths of the other 2 sides are as follows:
 - 9
 - 6
- A note indicates the figure is not drawn to scale.

In the right triangle shown, which of the following is closest to the value of c ?

- A. 7.5
- B. 10.8
- C. 15
- D. 58.5



The circle above with center O has a circumference of 36. What is the length of minor arc $\widehat{AC}$?

- A. 9
- B. 12
- C. 18
- D. 36

Q5

GRID-IN

GEOMETRY AND TRIGONOMETRY

Each side of a square has a length of 45. What is the perimeter of this square?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

GEOMETRY AND TRIGONOMETRY

In a right triangle, the measure of one of the acute angles is 51° . What is the measure, in degrees, of the other acute angle?

- A. 6
- B. 39
- C. 49
- D. 51

Q7

GEOMETRY AND TRIGONOMETRY

A right triangle has legs with lengths of 11 centimeters and 9 centimeters. What is the length of this triangle's hypotenuse, in centimeters?

- A. $\sqrt{40}$
- B. $\sqrt{202}$
- C. 20
- D. 202

Q8

GEOMETRY AND TRIGONOMETRY

What is the value of $\cos\frac{565\pi}{6}$?

A. $\frac{1}{2}$

B. 1

C. $\frac{\sqrt{3}}{2}$

D. $\sqrt{3}$

Q9

GRID-IN

GEOMETRY AND TRIGONOMETRY

A triangle has a base length of 40 centimeters and a height of 90 centimeters. What is the area, in square centimeters, of the triangle?

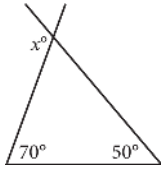
STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10

GEOMETRY AND TRIGONOMETRY



In the figure above, two sides of a triangle are extended. What is the value of x ?

- A. 110
- B. 120
- C. 130
- D. 140